

Implement a Generative Adversarial Network (GAN) to generate synthetic images or tabular data.

Dataset: MNIST (for images) or Synthetic Financial Transactions.

Code:

```
import torch
import torch.nn as nn
from torchvision import datasets, transforms
from torchvision.utils import save_image
from torch.utils.data import DataLoader

# Load dataset
transform = transforms.Compose([transforms.ToTensor(),
                                transforms.Normalize([0.5], [0.5])])
loader = DataLoader(datasets.MNIST('.', train=True, download=True,
                                   transform=transform), batch_size=64, shuffle=True)

# Generator
G = nn.Sequential(
    nn.Linear(100, 256), nn.ReLU(),
    nn.Linear(256, 512), nn.ReLU(),
    nn.Linear(512, 784), nn.Tanh()
)

# Discriminator
D = nn.Sequential(
    nn.Linear(784, 512), nn.LeakyReLU(0.2),
    nn.Linear(512, 256), nn.LeakyReLU(0.2),
    nn.Linear(256, 1), nn.Sigmoid()
)

loss = nn.BCELoss()
optG = torch.optim.Adam(G.parameters(), 0.0002)
optD = torch.optim.Adam(D.parameters(), 0.0002)

# Training
for epoch in range(10):
    for real, _ in loader:
```

```

real = real.view(-1,784)

# Train Discriminator
z = torch.randn(real.size(0),100)
fake = G(z)

d_loss = loss(D(real), torch.ones(real.size(0),1)) + \
          loss(D(fake.detach()), torch.zeros(real.size(0),1))

optD.zero_grad()
d_loss.backward()
optD.step()

# Train Generator
z = torch.randn(real.size(0),100)
fake = G(z)

g_loss = loss(D(fake), torch.ones(real.size(0),1))

optG.zero_grad()
g_loss.backward()
optG.step()

print("Epoch:", epoch, "D Loss:", d_loss.item(), "G Loss:",
g_loss.item())

# Generate images
z = torch.randn(25,100)
images = G(z).view(-1,1,28,28)
save_image(images,"generated.png",nrow=5,normalize=True)

```

Output:

```

100%|██████████| 9.91M/9.91M [00:00<00:00, 57.7MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 2.73MB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 14.6MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 8.23MB/s]
Epoch: 0 D Loss: 0.10673831403255463 G Loss: 6.451478004455566
Epoch: 1 D Loss: 0.33933961391448975 G Loss: 2.919437885284424
Epoch: 2 D Loss: 0.31025588512420654 G Loss: 5.303173542022705
Epoch: 3 D Loss: 0.18195413053035736 G Loss: 7.916196346282959
Epoch: 4 D Loss: 0.4277549088001251 G Loss: 5.050975799560547
Epoch: 5 D Loss: 0.11766433715820312 G Loss: 5.152519702911377
Epoch: 6 D Loss: 0.21371516585350037 G Loss: 6.847424507141113
Epoch: 7 D Loss: 0.5394768714904785 G Loss: 5.32853364944458
Epoch: 8 D Loss: 0.37341782450675964 G Loss: 6.850861549377441
Epoch: 9 D Loss: 0.2031562775373459 G Loss: 4.2493109703063965

```